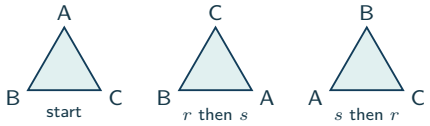
ALGEBRA LADDER 6 OF 9

Turning and flipping a triangle

Six moves of a triangle form a small world where the product of two moves is another move and the order matters.

1 / THE RULE AND THE PICTURE

Turn by a third of a circle and flip across the vertical, and the order changes the result



Let r turn the triangle by $120°$ anticlockwise and s flip it left to right. The symbols sr and rs mean the move made by doing the right one first, as with functions, and

$$sr \neq rs$$

because the labelled corners end in different places, though both products are again one of the six moves.

2 / THE TABLE OF ALL PRODUCTS

Every row and column contains each of the six moves exactly once

	e	r	r^2	s	sr	sr^2
e	e	r	r^2	s	sr	sr^2
r	r	r^2	e	sr^2	s	sr
r^2	r^2	e	r	sr	sr^2	s
s	s	sr	sr^2	e	r	r^2
sr	sr	sr^2	s	r^2	e	r
sr^2	sr^2	s	sr	r	r^2	e

row move done after column move

The six moves are doing nothing (e), the turns r and r^2 ($240°$), and the three flips s, sr and sr^2. Reading the table, $sr = r^2 s$ and $r^3 = e$ and $s^2 = e$, so a flip undoes itself and the turn r is undone by r^2.
The table is a Latin square, which is the footprint of inverses: to get any move from any other there is exactly one move to apply.

3 / THE LAWS THAT HOLD

Closed, associative, with identity and inverses

Every product of two moves is a move, so the set is closed, and doing nothing is the identity. Every move has an inverse, since $r^{-1} = r^2$ and each flip is its own inverse. Associativity holds for all 216 triples of moves, and it holds in general because moves are composed like functions. A set with these three properties is a group, and this one is called D_3.

4 / WHO COMMUTES WITH WHOM

Exactly half of the ordered pairs commute

Of the 36 ordered pairs of moves exactly 18 commute: the three turns e, r, r^2 commute with each other, as flat rotations always do, and each move commutes with itself and with the identity, but a flip and a turn never commute unless the turn is the identity, and two different flips do not commute either.
The rule $sr = r^{-1}s$ says that passing a flip across a turn reverses the turn.

5 / TRY THIS

Cut out a paper triangle and label the corners

Colour the corners A, B, C and apply r then s, then start again with s then r, and compare the corner order with the picture.
A second task is to find which single move undoes sr: since $(sr)(sr) = e$ the move sr is its own inverse, because it is a flip, so doing it twice returns the triangle.

ASIDE / THE SMALLEST NON-COMMUTATIVE GROUP

With six elements this is the smallest group where order matters, and the same group is the six ways of arranging three objects.

LINKS TO OTHER SHEETS

The turns alone behave like clock arithmetic in the number theory ladder, and the next sheet writes turns and flips as matrices.

Convention. e is doing nothing, r a turn of $120°$ anticlockwise, s a left to right flip; sr means r first and then s; D_3 is the dihedral group of the triangle.